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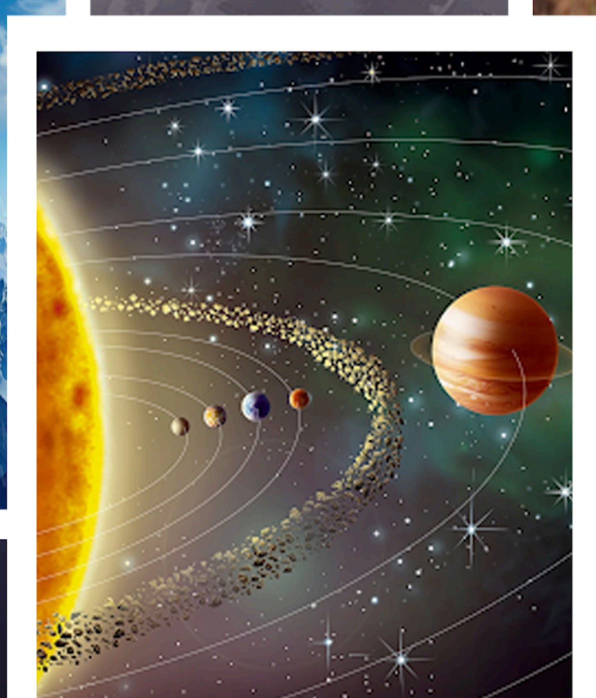
General Knowledge

World Geography

STUDY NOTES

Latitudes & Longitudes



Latitude & Longitudes

Axis of Earth

- Earth's rotational needle-like axis is an imaginary line that runs through the North and South Pole.
- The earth moves around this needle from **west to east.**

Equator

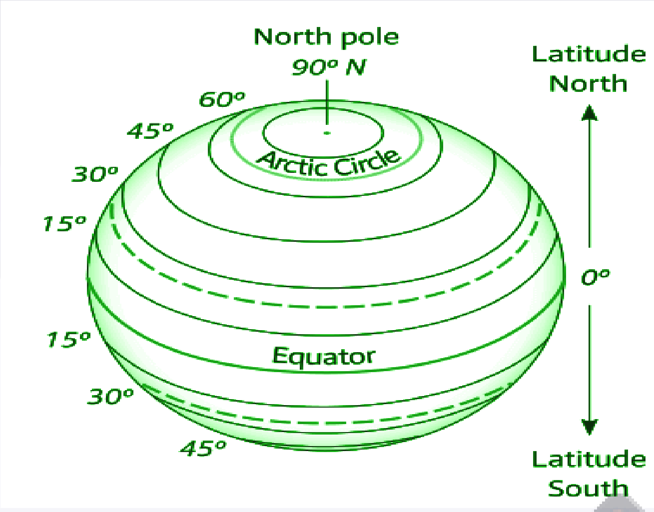
- An imaginary line running on the globe divides it into two equal parts is called the **Equator.**
- The Northern half of the Earth is called the **Northern Hemisphere** and the southern half is called the **Southern Hemisphere.**
- All parallel circles from the equator up to the poles are called parallels of latitudes, measured in degrees.
- The Equator represents the **zero-degree latitude.**
- All parallel north of the equator are called **'north latitudes'** and parallel south of the equator are called **'south latitudes'**.
- The size of the parallels of latitude decreases as we move away from the equator.

LATITUDES

- Lines joining the places with the same latitudes are called **parallels**.
- The value of the Equator is **0°** and the latitude of the poles are **90° N and 90° S.**
- The total number of parallels thus drawn, including the equator, **will be 179.**
- If we included the poles as full latitude lines, the count would become 181, but typically they are considered points instead.
- **At the equator, it is 110.6 km and at the poles, it is 111.7 km.**

Important Parallels of Latitudes

- Besides the equator (0°), the North Pole (90°N), and the South Pole (90°S) there are four important parallels of latitudes-



➢ **Tropic of Cancer (23½°N)** in the Northern Hemisphere

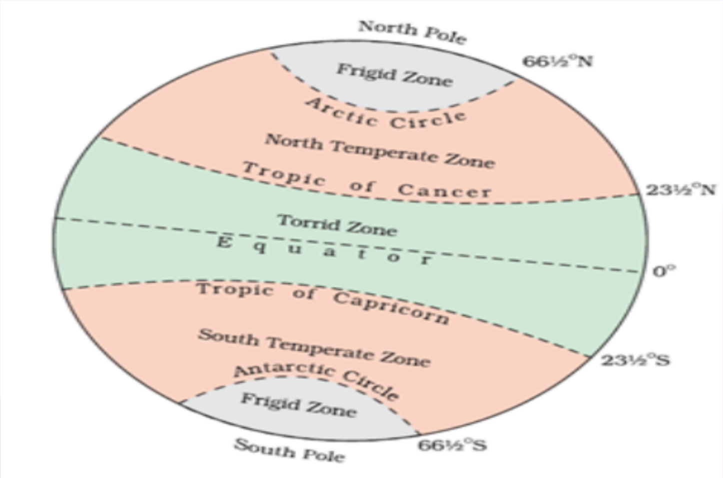
➢ **Tropic of Capricorn (23½°S)**in the Southern Hemisphere

➢ **Arctic Circle at (66½°)** north of the equator.

➢ **Antarctic Circle at (66½°)** south of the equator.

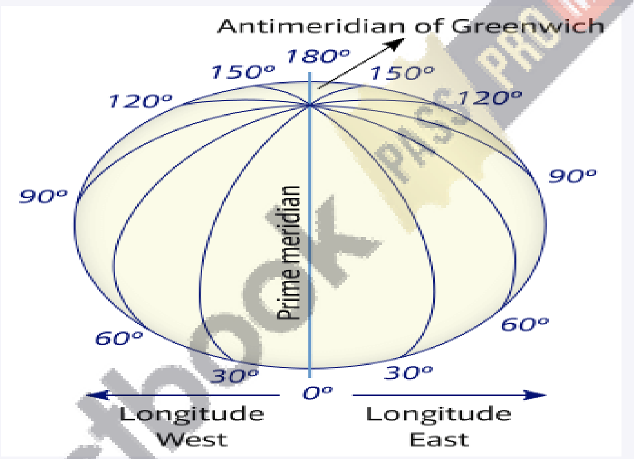
Latitudinal Heat Zones of the Earth

- **The Torrid Zone** refers to the region of the Earth located **between the Tropic of Cancer (23½° N) and the Tropic of Capricorn (23½° S)**. It is the hottest climatic zone, receiving direct sunlight throughout the year, leading to consistently high temperatures.
- **The Temperate Zone** lies **between the Tropic of Cancer (23½° N) and the Arctic Circle (66½° N)** in the Northern Hemisphere and **between the Tropic of Capricorn (23½° S) and the Antarctic Circle (66½° S)** in the Southern Hemisphere.
- **The Frigid Zones** are the areas north of the Arctic Circle and south of the Antarctic Circle.



LONGITUDES

- Longitude refers to the geographic coordinate that specifies the **east-west position** of a point on the Earth's surface.
- These lines of reference are called the meridians of longitude with each degree being further divided into minutes and seconds.
- They are semicircles and the distance between them decreases steadily towards polewards until it become **zero at the poles.**
- The Prime Meridian passes through Greenwich, England, and divides the Earth into the Eastern Hemisphere and Western Hemisphere. It is designated as 0° longitude.
- There are **360 lines of longitude—180° east and 180° west of the Prime Meridian**.

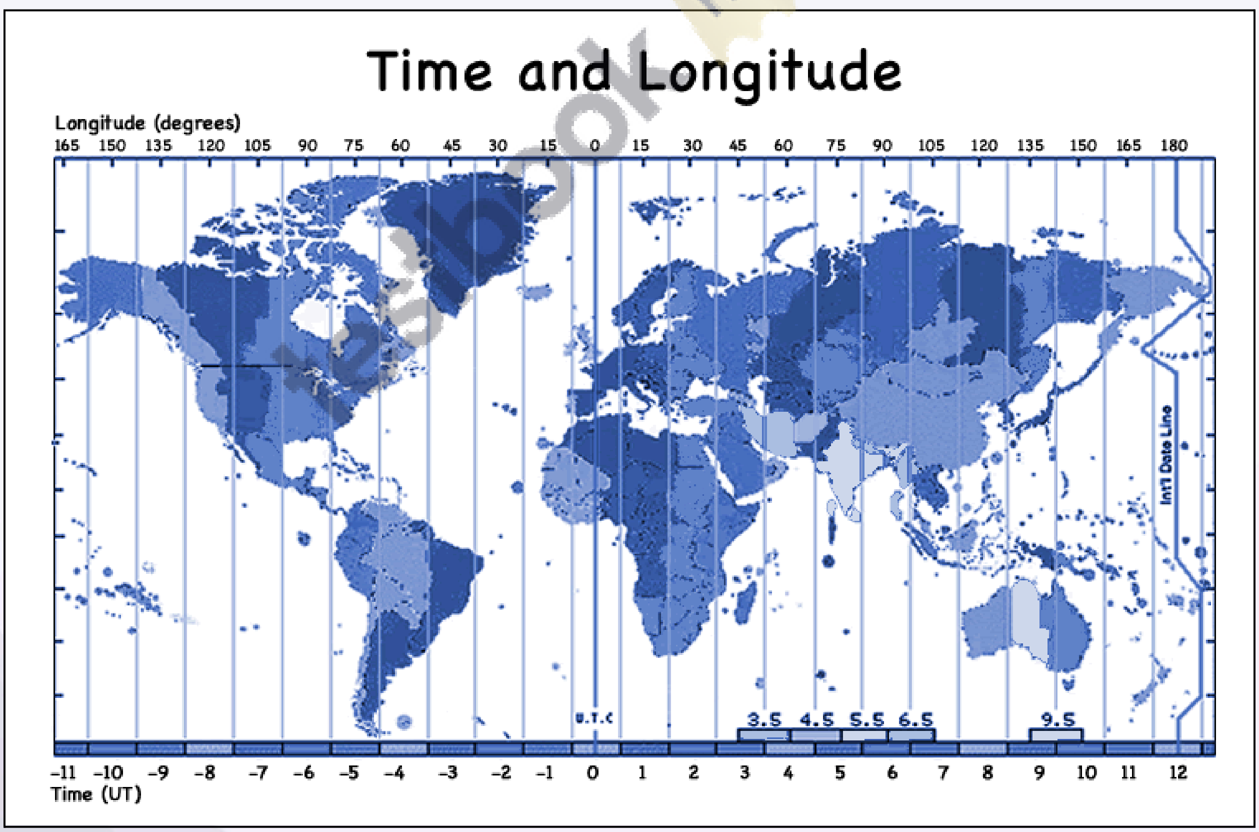


TERM	LATITUDE	LONGITUDE
Direction	East-west parallel to the equator	North-South: converging at the poles and widest at the equator
Parallel lines	Yes	No
Range	0 to 90 degrees North and South	0 to 180 degrees East and West
Denoted by	Greek letter phi(ϕ)	Greek letter lambda (λ)
Divides Hemisphere	Northern and Southern	Eastern and Western

Denotes distance from	Equator(north or south)	Prime Meridian (east or west)
Time zone	Locations that share the same latitude do not necessarily fall in the same time zone.	All locations on the same longitude fall in the same time zone
No. of lines	179	360
Notable lines	Equator, Tropic of Cancer, Tropic of Capricornn	Prime Meridian, International date line
Applications	Classifying temperature zones	Classifying time zone

- The point of intersection of Latitude and Longitude gives the location of a place on the globe.

Longitude and Time



- The movement of the Earth, the Moon, and the Planets can measure time.

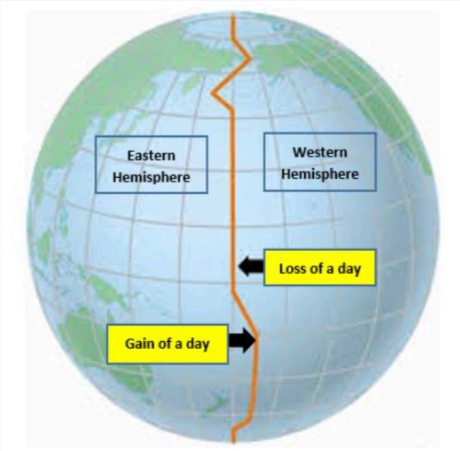
- As the Earth rotates from west to east, those places east of Greenwich will be ahead of Greenwich time and those to the west will be behind it.
- The rate of difference can be calculated as follows: The earth rotates 360° in about 24 hours, which means **15° in an hour or 1 in four minutes.** Thus when it is noon at Greenwich, the time at 15° east of Greenwich will be 15x4=60 minutes, i.e., 1 hour ahead of Greenwich time which means 1 P.M.
- But at 15° west of Greenwich, the time will be behind Greenwich time by one hour, i.e., it will be 11.00 a.m.

Standard Time

- The local times and places of different meridians are bound to differ.
- In India, there will be a difference of about 1 hour and 45 minutes in the local time of Dwarka in Gujrat and Dibrugarh in Assam.
- **In India, 82½° E (82° 30" E) is treated as the standard meridian and known as Indian Standard Time(IST)**
- **India is located 82° 30" E is 5 hours and 30 minutes ahead of Greenwich's mean time**. So it will be 7:30 pm in India when it is 2:00 pm in London.

International Date Line

- The line from where it is considered that day truly starts is called the International Date Line **180° longitude** from where this line passes.
- The time at this longitude is **exactly 12 hours from the 0° longitude**, irrespective of whether one travels in the east or west direction.



Some interesting facts about standard time

- Some countries have great longitudinal extent so they have adopted more than one standard time.
- For example, Russia has 11 standard times.
- **France** holds the record for the most time zones, with a total of **12 time zones**
- The Earth has been divided into 24 time zones of one hour each.
- Each zone thus covers 15° longitude